

Pediatric Dermatoses

Aka: My kid has a rash!

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Objectives

- Know the typical appearance/treatment of **common infant rashes**.
- Know clinical features/treatment of **seborrheic dermatitis** (cradle cap).
- Know the common bacterial causes, typical clinical appearance, and treatment of **impetigo**.
- Know the organism that causes **molluscum contagiosum** and describe their clinical appearance and management.
- Visually identify the common lesions seen in **tinea capitis** and **tinea corporis** and know the difference in treatment between them.
- Know the common anatomic distribution of **lice and scabies infestation**, and treatment.
- Know the presentation, treatment, and outcome of **common congenital neoplasms**.
- Understand the prevention of, and describe the two common types of **diaper rash** (**irritant contact dermatitis** and **candidal dermatitis**) by their etiology, clinical presentation, and treatment.
- Know the causes of three different **hypersensitivity syndromes**.

Not covered here:

- **Most dermatologic manifestations of systemic illness**
(with exception of selected emergencies)
- Strep/scarlatina, meningococemia, staph (including SSS)
- Viral exanthem
- Bacterial infections
- Urticarial lesions
- Vasculitidies
 - Henoch-Schonlein Purpura
 - Kawasaki Disease
 - Etc.

Hematologic conditions:

- Idiopathic Thrombocytopenic Purpura
- Petechial/purpuric rashes in general
- For the most part we cover **chief complaint: rash**

Recommended Reading

This lecture powerpoint is the only required reading.

UpToDate: Benign skin and scalp lesions in the newborn and young infant

AAFP articles

Newborn Skin: Part I. Common Rashes

Am Fam Physician. 2008 Jan 1;77(1):47-52

<http://www.aafp.org/afp/2008/0101/p47.html>

Newborn Skin: Part II. Birthmarks

Am Fam Physician. 2008 Jan 1;77(1):56-60

<http://www.aafp.org/afp/2008/0101/p56.html>

Web references

www.dermatlas.net

www.dermnetnz.org

<https://www.jaad.org/collection-skin-of-color-images>

General Principles

- It's not all about the exam! **History is KEY**
 - Age, time of onset, duration, past and family history, social history and environment, and medications.
 - Can often elicit the diagnosis before even looking at the skin.
- **Expose the entire surface** of the skin.
- Being able to **describe** the lesion is important:
 - Distribution, configuration, color, secondary changes, primary changes. (“generalized, discrete, red, scaly papules.”)
- **Reassurance**—treat the child, not the anxious parent.
- Keep **treatment** simple.
 - Reassurance is often all that's needed.
 - Usually topical > systemic (some expectations will be discussed)

Neonatal Rashes



Erythema Toxicum Neonatorum

- Self-limited, benign
- **Common:** up to 60% of term infants (less common in LBW/premature infants)
- Etiology unknown (? immaturity of pilosebaceous follicles)
- Erythematous macules surrounding small pustules (containing eosinophils) “Flea bites”
- Can occur anywhere (most commonly face, arms, buttocks, and torso), spares palms/soles



Erythema Toxicum Neonatorum

- No signs of systemic illness (fever, lethargy, poor feeding)
- DDx:
 - Bacterial infection (*S. Aureus*, GBS)
 - Fungal infection (candida)
 - Viral infection (HSV, varicella)
 - Transient neonatal pustular melanosis
- Onset: first 1-4 days (not present at birth).
- Duration: 2-3 weeks.
- Tx: reassurance.



Transient Neonatal Pustular Melanosis

- Self-limited, benign
- Pustular lesions (PMN's); rupture easily, can involve palms/soles
- No erythematous base
- After rupture evolve into hyperpigmented macules
- Onset: Birth or first 24 hours of life
- Duration: 3 wks – 3 mo
- Tx: reassurance



Transient Neonatal Pustular Melanosis

● DDx:

- Erythema toxicum
- Bacterial infection
- Viral infection
- Fungal infection
- Milia
- *Infantile acropustulosis*
(crops of pruritic vesiculopustules
(palms/soles/limbs), presents at
birth or 1st year, resolves in 2 years
(tx topical steroids))



Cutis Marmorata (physiologic)

- Transient, benign mottling of skin; resolves with warmth.
- Possibly exaggerated vasomotor response to decreased core temp.
- Blanching mottled or lacy erythema on trunk and extremities.
- Tx: Reassurance, rewarming



Milia

- Up to 50% of newborns.
- Small 1-2mm superficial keratin cyst forming nonconfluent pearly papules with minimal surrounding erythema; typically on **face, eyes, nose**; occasionally trunk.
- Usually disappear in 2-4 weeks, may have recurrences.
- DDx:
 - Pustular melanosis, seborrheic dermatitis.
- Tx: Reassurance



Miliaria

- **Sweat retention** when eccrine glands partially close □ sweat infiltration.
- Typically in intertriginous or occluded areas.
- Miliaria crystallina
 - Stratum corneum
 - Small non-inflammatory vesicles, that ruptures and desquamates within days.
- Miliaria rubra
 - Sub-epidermal
 - **Heat rash**
 - Pruritic erythematous macules and vesicles on warm skin.
- Tx: Reassurance; avoid occlusion; cool environment



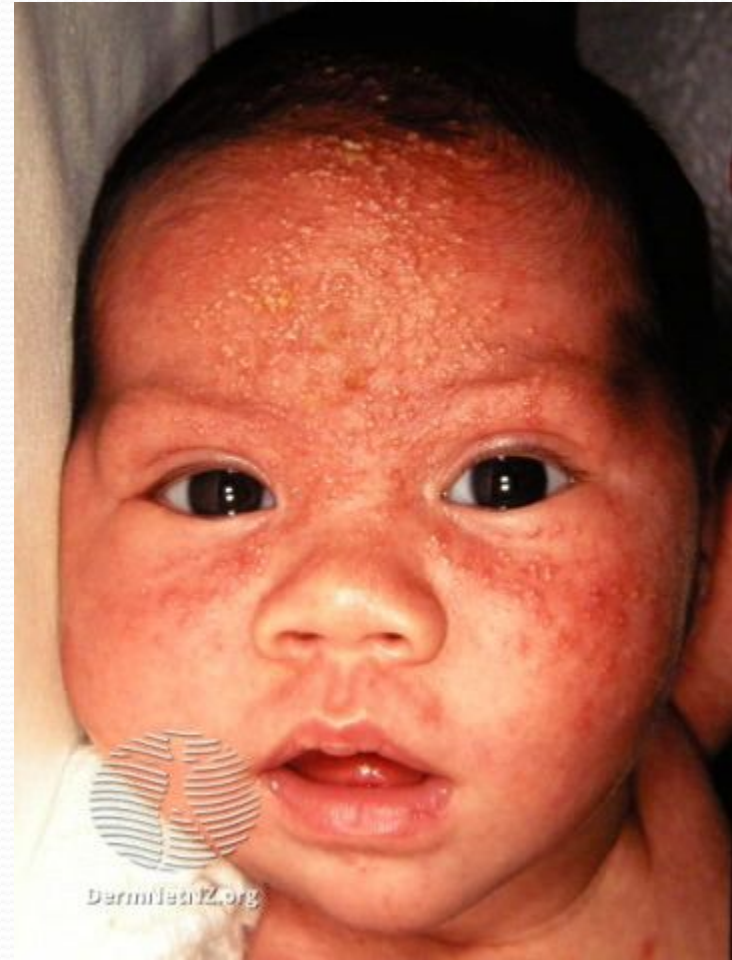
Acne Neonatorum

- Acneiform eruption (comedones, papules, pustules) on forehead, cheeks, and nose in newborns and infants.
- Pearly papules with surrounding erythema.
- Possibly due to heightened sensitivity of skin due to maternal hormones.
- Peaks around 2 months. Wanes with maternal hormones around 3-4 months (without scarring).
- DDx: other common neonatal skin findings, infection, -hyperandrogenism (if severe and other signs, consider CAH or virilizing tumors)
- Tx: Reassurance. If extensive can consider 2.5% benzoyl peroxide

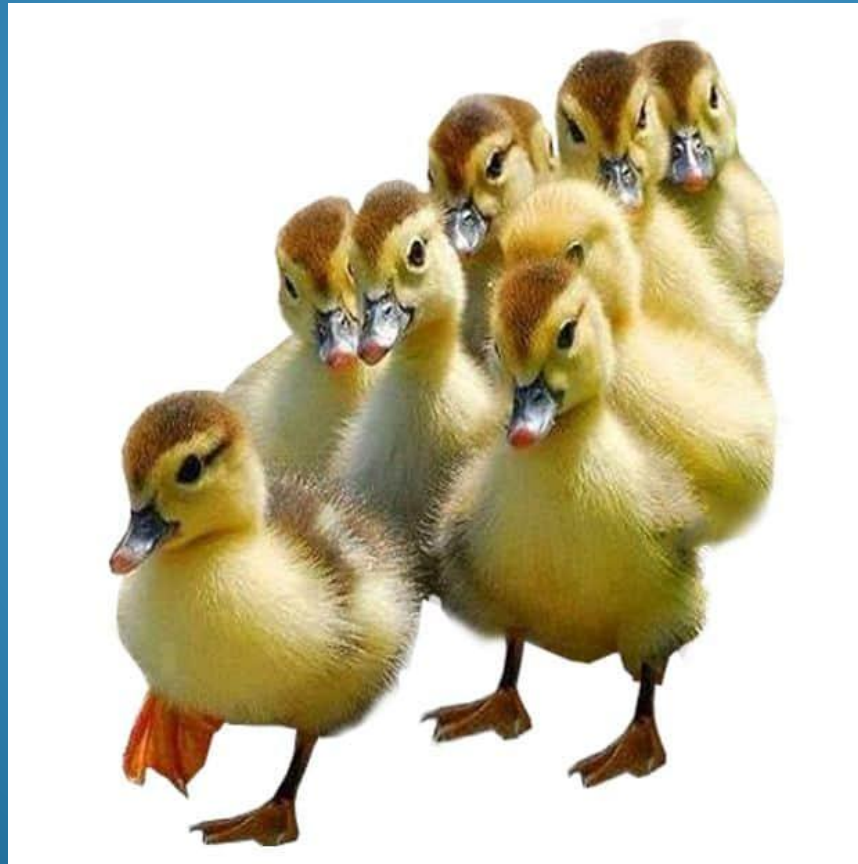


Neonatal Cephalic Pustulosis

- variant of neonatal acne
- pustular eruption on face and/or scalp of newborns (around week 3)
- Unlike true neonatal acne, there are no **comedones** (blackheads, whiteheads).
- related to **malassezia** colonisation, which may be found on **mycology** (microscopy and culture) of smears from facial pustules.
- generally resolves without treatment.
- **topical antifungal** (eg ketoconazole) safe/can be helpful (image: DermnetNZ)



Congenital Lesions



Slate Gray Patch

- AKA Congenital Dermal Melanocytosis
- Large **benign** bluish-gray/black pigmented nevus of lower back, buttocks, thighs.
- More common in black, native american, asian, and hispanic infants.
- Most start to fade by 2 years old and many completely disappear between 7 and 12 years.
- DDx: ecchymosis, non-accidental trauma
- Tx: Reassurance, **good documentation.**



Capillary Malformation

Nevus Simplex

- AKA stork bite/angel's kiss, salmon patches
- Present in some form in almost all newborns
- **Present at birth**, most resolve by 18 months of age
- May darken when child cries
- Tx: observation



Capillary Malformation

Nevus Flammeus

- AKA Port Wine stain
- Collection of dilated mature capillaries
- **Present at birth**, some persist throughout life
- Pink to purple macular lesion
- May darken when child cries
- **Do not regress**
- Associations:
 - **Sturge Weber Syndrome**
 - Klippel Trenaunay Weber
 - V₁ Distribution of Trigeminal nerve → ipsilateral glaucoma (refer to Ophtho)
- **Tx: observation; pulsed dye laser** (lighten, ideally before 1 yr)



Hemangiomas

- Vascular tumor of proliferating immature capillaries (1-2% of newborns)
- Not usually present at birth
- **Note the distribution**
- Expand during first 1-2 years, then involute
 - 50% by 5 years
 - 90% by 9 years
- Reddish-purple raised nodule on skin
- Tx:
 - **Observation** – usual treatment of choice (depending on location!)
 - Systemic propranolol
 - Steroids (topical, systemic, intralesional)
 - Pulse-dye laser
 - Surgery



Eczematoid Eruptions

Eczematoid Eruptions

- A group of disorders characterized by pruritus and cutaneous reactivity (“the itch that rashes”)
- Describes a complex of symptoms:
 - Acute phase: itching, burning, redness, papules, vesicles, and crusting
 - Chronic phase: thickened plaques with lichenification (hypertrophied dermis from chronic scratching/rubbing ->thick leathery skin) and scaling



Atopic Dermatitis

- Diagnosis
 - History and Physical (pruritis, morphology and distribution, chronic/recurring)
- Treatment
 - Bland Emollients
 - Short baths
 - Mild soaps
 - Topical steroids
 - No fragranced products
- Complications
 - Superinfection
 - Eczema Herpeticum (HSV, treat with antivirals, can be life threatening, vision threatening)



Pityriasis Alba (white patches)

-common skin condition

- 1) red, scaly patches that resolve, leaving
- 2) areas of scaling hypo-pigmentation/ lighter coloration (usually the time treatment is sought)

-cause unknown, ? related to atopic dermatitis

-round, oval or irregular in shape and red, pink or skin colored.

-Several patches may be observed at once usually on the face and arms.

-variable duration (months to over a year)

-Can be recurrent

-Dx: usually clinical

Tx: gentle skin care, sun protection, moisturizer, consider topical steroid if inflamed/flaky

Prognosis: often self resolves, more noticeable if surrounding skin is exposed to sun and tans (eg summer break!). Loss of pigment is not permanent.



Diaper Dermatitis

- **Most commonly irritant/allergic contact dermatitis.**
 - Due to mechanical irritation from diaper material and skin breakdown from exposure to urine and feces.
- **Second most common is candidal dermatitis.**
 - Usually secondary to skin breakdown from irritant contact dermatitis. May occur following antibiotic use.



CANDIDA DERMATITIS

Diaper Dermatitis

- Irritant/allergic contact dermatitis.
 - Confluent patches of scaly erythema.
 - **Skin folds spared (not exposed to the trigger)**
- Candidal dermatitis.
 - Confluent patches of scaly erythema.
 - **Beefy** erythema.
 - **Skin folds involved, satellite lesions.**
 - *Consider if failed therapy for contact dermatitis.*



Diaper Dermatitis Treatment

Irritant/Allergic contact dermatitis.

- Treat with **barrier cream (eg: zinc oxide)** and frequent diaper changes.
- No wipes or fragrances (clean with water)
- If recent change in diaper, consider going back to old choice
- “Diaper holidays”, keep dry and clean

Candidal dermatitis.

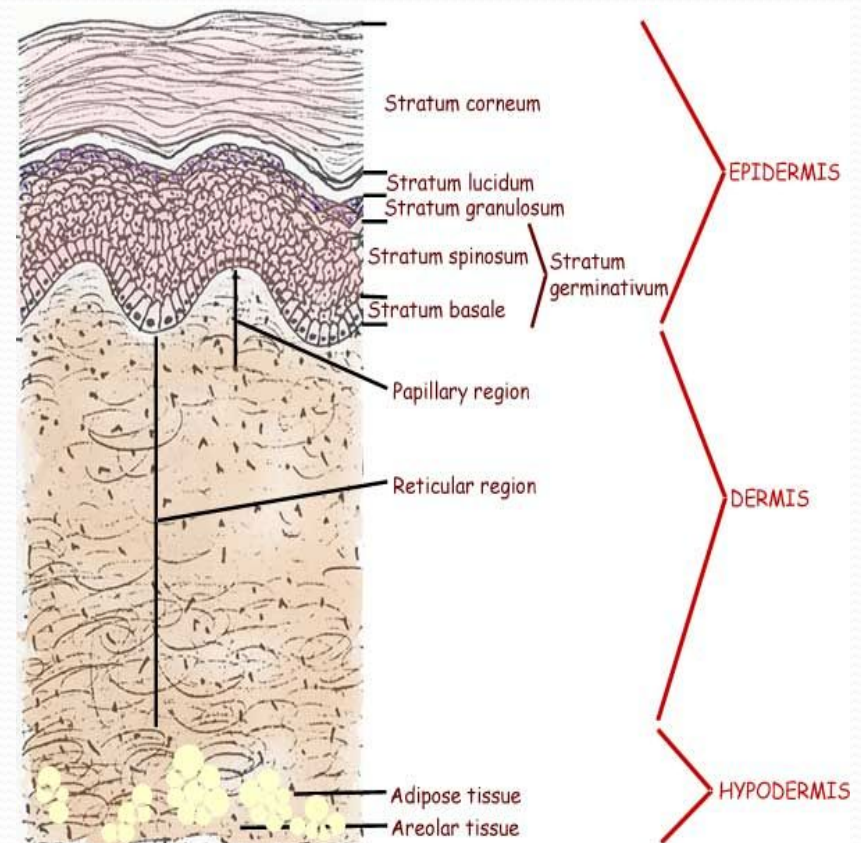
- Add antifungal cream (nystatin, clotrimazole)

Papulosquamous Disorders



Papulosquamous Disorders

- Abnormal maturation (stratum spinosum)
 - Pityriasis rosea
- Increased production (stratum basale)
 - Psoriasis (explored elsewhere)
 - Seborrheic dermatitis
- Increased adherence (stratum corneum)
 - Ichthyosis



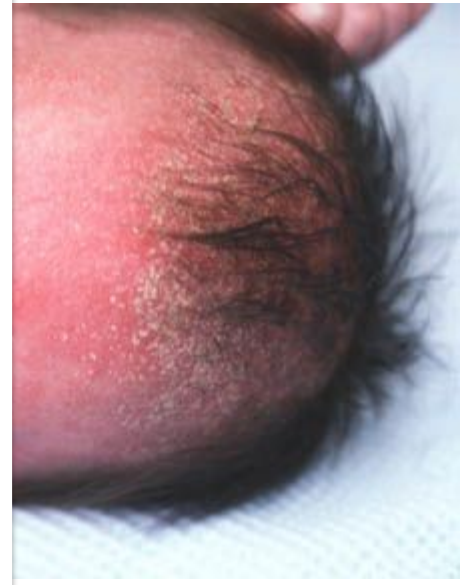
Pityriasis Rosea

- Characteristic “Herald Patch” (2-5cm) with “collarette of scale” followed by patches in a “Christmas Tree” distribution
- Herald patch can be confused with fungal
- Possible viral etiology (HHV-7)?
- May have pruritus
 - Topical steroids
- Self resolves



Seborrheic Dermatitis

- Common (*cradle cap*)
- Chronic disorder marked by redness and scaling in areas with dense concentrations of **sebaceous glands** (scalp, eyebrows, body folds)
- Fine *flaky to thick greasy yellow scales* with or without an underlying erythematous patch or plaque base.
- ?*Malassezia furfur*
- Usually resolves by one year old.
- Tx:
 - Manual removal of scales
 - Medicated shampoos (tar, selenium, ketoconazole)
 - Topical corticosteroids



Terra-firma Forme Dermatosis



- raised, dirty-appearing areas of skin (terra firma = dry land in Latin).

- disorder of keratinization = abnormal/delayed development of skin cells (prolonged adhesion -->. retention of melanin, sebum, and microorganisms)*



- mainly affects children and adolescents

- thickened, brown-black papules and plaques on the neck, trunk, face, and abdomen including the umbilicus. The plaques can have a papillomatous surface.

- Psychosocial distress can result from the dirty appearance which may be perceived to reflect poor personal hygiene.

Terra-firma Forme Dermatosis

DDx:

Dermatitis neglecta: inadequate washing (resolved with soap and water)

Acanthosis nigricans,

Confluent and reticulated papillomatosis of Gougerot and Carteaud,

Atopic dirty neck, is a dirty-appearing acquired hyperpigmentation in setting of atopic dermatitis

Symmetrical acrokeratoderma, presents with dirt-like hyperkeratotic areas distributed symmetrically on the upper and lower limbs, particularly the wrists, ankles, and dorsa of the hands and feet [12]

Seborrhoeic keratosis, is a benign growth of keratinocytes, not resolved with alcohol wipe

Pityriasis versicolor

Epidermal naevus

Postinflammatory hyperpigmentation.

Tx:

Rubbing with ethyl or isopropyl alcohol-soaked wipes (diagnostic and therapeutic)

Mild topical [keratolytic agents](#) (eg, 5% [salicylic acid](#) in petrolatum ointment) in extensive cases

[Chemical peels](#) (eg, 20% salicylic acid in alcohol)

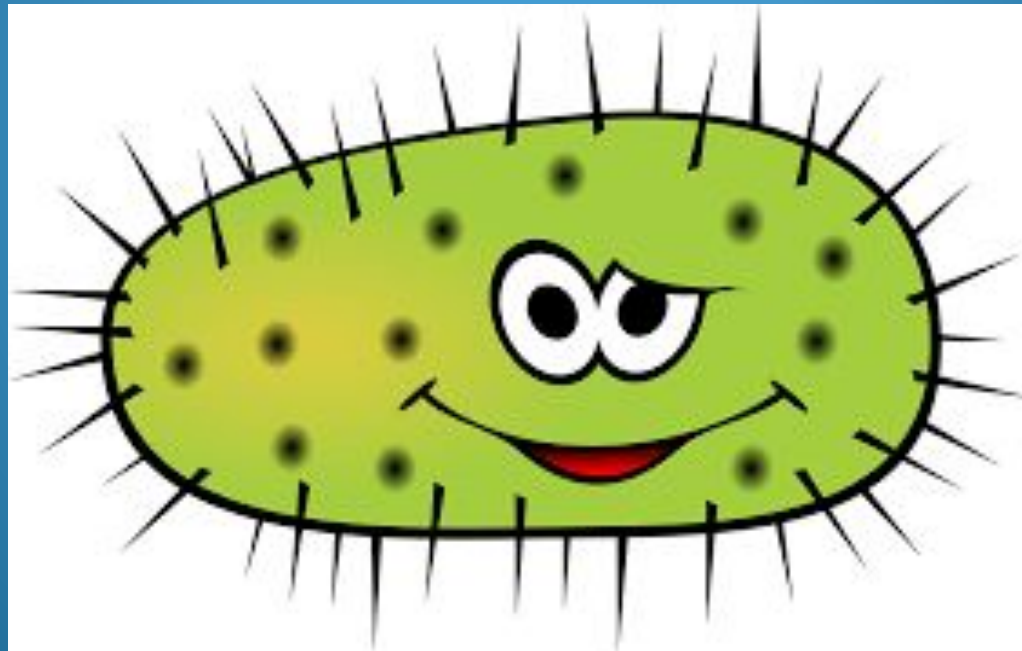
Tx: Reassurance -- benign, responds to treatment, rarely recurs. Alcohol wipes for recurrence

Ichthyosis

- Congenital disorders of keratinization leading to **dry scaly skin**
- **Ichthyosis vulgaris**
 - **Most common**
 - Autosomal dominant
 - **Onset in early childhood (normal at birth)**
 - Dry, rough scaly skin by age 5
 - Improve with age
- Sex linked Ichthyosis
 - X-linked recessive
 - Onset at or around birth
 - “Dirty” brown scales
 - Same or worse with age



Infectious Disorders



Molluscum Contagiosum

- Caused by a **pox virus** (*molluscum contagiosum*) - more in Viral Infections of the Skin lecture
- Flesh colored to pink dome-shaped **umbilicated papules**
- Usually seen ages 3-16 years (if extensive in adult consider HIV test)
- Tx:
 - **Observe** (usually resolves in 6-18 months)
 - Destructive techniques (liquid nitrogen, etc)
 - New tx NO₂ topical Berdazimer (2024)



Impetigo



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Impetigo

- Usually caused by **staph aureus** or **Group A strep**
- non-bullous or bullous (staph exfoliative toxin A)
- vesicles and pustules with **“honey colored crusting”**
- Tx: **Topical antibiotics for localized infection (Mupirocin)**
- Oral abx reserved for widespread or advanced infections (Cephalexin, Dicloxacillin, Bactrim, Clindamycin (MRSA))
- Sequelae: Post-strep GN



Tinea Corporis

- “Ringworm”: fungal
- Erythematous, pruritic, and scaly lesions usually with an annular shape that grows outward leaving a central clearing
- Tx: topical antifungal “-azoles”



Tinea Capitis

- **Fungal infection** of the scalp and hair follicle
- Broken off hairs w/scaling
 - “Black dot sign” (broken hairs)
 - Tricophyton-will not fluoresce
- Can form **kerion** (boggy/ tender, inflammation) - add steroid to decrease scarring/hair loss
- Antibiotics for superinfection
- Dx: KOH exam of spores/skin scraping
- DDx: alopecia areata, seborrheic dermatitis
- Tx: oral **griseofulvin** x 6-8 wks, **terbinafine** x 4 wks



Gianotti Crosti Syndrome (Papular Acrodermatitis of Childhood)

- papules on extremities/face/ears
- Weeks
- clusters knees/elbows
- "ID" reaction: 2ndary immunologic response post viral (EBV, molluscum?), also seen with eczema
- supportive care: topical steroids, antihistamines



Tinea Versicolor/Pityriasis Versicolor

- fungal (yeast) overgrowth
-malassezia species
- scaly macules/color changes to skin (hyper/hypo pigmentation, erythema, etc)
- scaling indicates active infection
- Tx: topical antifungal (Clotrimazole, ketoconazole, etc) vs. oral (if hard to reach)
- confused with vitiligo?

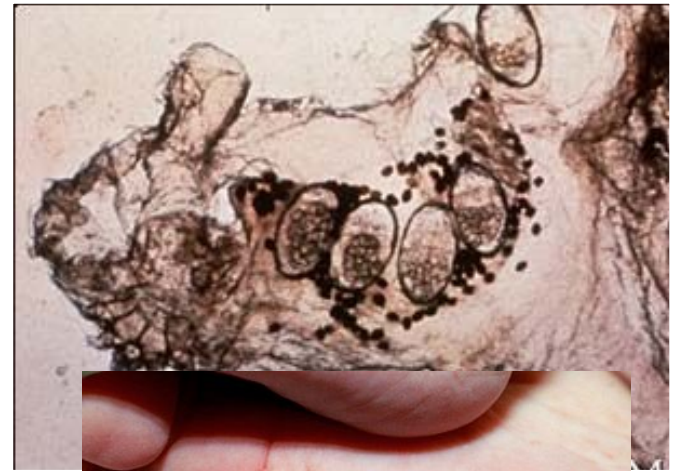


Infestations



Scabies

- Caused by *Sarcoptes scabiei* mite
- Highly contagious
- Extremely pruritic erythematous papules or scaly linear burrows in web spaces, groin, buttocks, axillae, arms, or trunk (younger babies look for head/neck, palms/soles)
- Diagnosed clinically or with skin scrapings looking for mites, eggs, or feces
- Tx: permethrin 5% cream for all household contacts, and cleaning of all clothing and linens.



Bed bugs (Cimex lectularius)



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- Usually a mild (but itchy) cutaneous reaction, but can lead to more severe response like anaphylaxis
- Immediate to 24 hrs (usually bite at night)
- About the size of an apple seed
- Tx: antihistamines, topical corticosteroids
- Eradication: difficult and extensive, high heat helps, resistant to pesticides and do not need to feed often
- Detecting difficult, (small and able to hide in tiny cracks and crevices.
- Evidence in bedding and on mattresses (dark brown excrement), old furniture

Lice (Pediculosis)

- Arthropods that infest the hair of the scalp (capitis), body (corporis), and pubic area (pubis).
- Attach to skin, feed on blood, lay eggs (nits) on hair shaft.
- Rarely, can be disease vectors (epidemic typhus, relapsing fever) – not head lice.
- **Lice infestation is contagious** (direct contact).
- May have intense itching, may see nits or lice. **Diagnosis may require repeated examinations.**
- Treatment: **Permethrin 1%**; Malathion for resistant infestations. Repeat treatment in one week. **Nit comb (wet comb systematically)**
- Nits may persist for months after tx



Hypersensitivity Syndromes



Erythema Multiforme

- “Erythema multiforme minor”
- Usually associated with **infection** (possibly medications)—*HSV*, *EBV*, *mycoplasma*
- Fixed, typically non-pruritic annular red macules, papules, and plaques symmetrically on trunk, extensor surfaces of arms and legs, palms and soles
- Tx: **supportive**, treat underlying infection



Id Reaction/Disseminated Secondary Eczema (Autosensitization/Autoeczematization)

-Localized inflammatory skin disease → Acute generalized dermatitis (cause unknown) ? immune response to skin component and/or to circulating infectious agents or cytokines (kids = adults)

Common preceding dermatitis types:

Chronic venous eczema

Acute contact dermatitis

Acute or chronic discoid eczema.

Associated Infections:

Fungal skin infection eg, inflammatory tinea pedis or kerion - dermatophytide

Bacterial skin infection (wound, burn)

Viral infection eg, molluscum contagiosum

Arthropod infestation eg, scabies or lice



Id Reaction

(Autosensitization/Autoeczematization)

Dx: acute, symmetrical, generalised acute eczema (pruritis disturbs sleep) - forearms, legs, thighs, trunk - varied appearance (blisters, bumps, crusted plaques (discoid eczema), follicular papules, morbilliform eruption, targetoid lesions, and vesicular hand dermatitis).

-Finding the cause is key! History and careful physical to look for inciting factor

Tx:

-Find and treat cause of primary rash (eg: systemic antibiotic/antifungal)

-Secondary eczema can be extensive and very pruritic/symptomatic

Wet wraps or dressings for weepy eczematous plaques

Potassium permanganate 1:10,000 soaks for localised oozing, infected areas

Potent topical steroid creams for 1-3 weeks

Systemic corticosteroid x several weeks

Oral sedating antihistamine at night

Derm referral (even possible admission)

Toxicodendron Contact

- Type IV hypersensitivity reaction to urushiol
- At least 50% of people react to as little as 1 ng
- Avoidance, washing with soap and water, and use of a barrier (Ivy Block)
- Systemic > local corticosteroid therapy



Toxicodendron Contact



Poison ivy



Poison oak



Poison sumac

**What Poison Ivy and
Poison Ivy Rash
Look Like**



Poison ivy rash

Photos: Courtesy of the American Academy of Dermatology

Phytophotodermatitis

-Form of **plant dermatitis**: natural photosensitising chemicals (furanocoumarins) + sunlight/UV exposure

24-48hrs later = phototoxic reaction

-Erythema/blistering: **varied patterns** depending on the way the exposure occurred (applied, eaten, drips, touched, etc)

-May ulcerate, scale → heal after several weeks to leave postinflammatory pigmentation (may be more intense in skin of color)

-Can also appear on exposed sites of the face, hands, and arms after eating large amounts of furocoumarin containing foods



Phytophotodermatitis

	Botanical name	Common names
Umbelliferae	Ammi majus	Queen Anne's Lace
	Heracleum spondylium	<u>Cow parsnip</u>
	Heracleum mantegazzianum	<u>Giant hogweed</u>
	Pastinaca sativa	<u>Parsnip</u>
	Apium graveolens	<u>Celery</u>
Rutaceae	Citrus bergamia	<u>Lime</u>
	Dictamnus albus	Gas plant, burning bush of Moses
Moracea	Ficus carica	<u>Fig</u> (from sap, not fruit)

Phytophotodermatitis

Dx: History of exposure + clinical appearance

DDx:

Allergic contact dermatitis to plants

Burns

Non-accidental injury

Dermatitis artefacta

Shiitake flagellate dermatitis

Complications:

Blistering

Ulceration/scarring

Infection

Postinflammatory hyperpigmentation

Tx:

Cool compresses

Topical antiseptics to prevent infection

Topical steroids

Analgesia

Avoid the inciting agents

Protective gear if part of work

Sunscreen/cover skin/avoid further sun exposure



The End